Detailed Lecture Notes on Lie Groups and Lie Algebras

Weeks 1 and 2: Summary

Introduction

These notes provide an introduction to the fundamental concepts of Lie groups and Lie algebras. Our primary goal is to understand their definitions, see key examples, and establish the crucial link between a Lie group and its corresponding Lie algebra. This relationship allows us to use the tools of linear algebra to study the structure and representations of these geometric objects.

1 Lecture 7: Lie Groups

1.1 Definitions

Definition 1.1. A **Lie group** is a set G that is simultaneously a group and a smooth manifold, such that the group operations are smooth maps. Specifically:

- The multiplication map $\mu : G \times G \to G$, given by $(g, h) \mapsto gh$, is smooth.
- The inversion map $i : G \to G$, given by $g \mapsto g^{-1}$, is smooth.

These two conditions are equivalent to the single requirement that the map $(g, h) \mapsto gh^{-1}$ is smooth.

- A **real Lie group** is based on a real smooth manifold.
- A **complex Lie group** is based on a complex analytic manifold with holomorphic group operations.

A **morphism** of Lie groups is a map that is both a group homomorphism and a smooth (or holomorphic) map of manifolds. A **representation** of a Lie group G is a morphism $\rho : G \to GL(V)$ for some vector space V.

When dealing with subgroups, a topological distinction is necessary.

Definition 1.2. A **Lie subgroup** (or **closed Lie subgroup**) H of a Lie group G is a subgroup of G that is also a closed submanifold. An **immersed subgroup** is the image of an injective morphism $\phi : H \to G$, which is an immersion but not necessarily a topological embedding.

Example 1.3 (An Immersed Subgroup)**.** A classic example is the torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$. Consider a line $L \subset \mathbb{R}^2$ with irrational slope. Its image in T^2 is a dense subgroup that is locally a 1-dimensional manifold but is not a closed submanifold. This is an immersed subgroup.

1.2 Examples of Lie Groups

Most fundamental examples of Lie groups are matrix groups, which are closed subgroups of the **general linear group** $GL_n(\mathbb{K})$, where $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$. $GL_n(\mathbb{K})$ consists of invertible $n \times n$ matrices and is itself a Lie group, being an open subset of the vector space of all $n \times n$ matrices.

Example 1.4 (**The Classical Groups**)**.**

- **Special Linear Group ($SL_n(\mathbb{K})$):** Subgroup of $GL_n(\mathbb{K})$ consisting of matrices with determinant 1. It is defined by the polynomial equation $\det(A) - 1 = 0$.
- **Orthogonal Group ($O_n(\mathbb{K})$):** Subgroup of $GL_n(\mathbb{K})$ preserving a non-degenerate symmetric bilinear form. With the standard form (the dot product), this is the group of matrices A satisfying $A^T A = I$. The **special orthogonal group** $SO_n(\mathbb{K})$ is the subgroup of O_n with $\det(A) = 1$. The group O_n is disconnected, while SO_n is connected.

- **Symplectic Group ($Sp_{2n}(\mathbb{K})$):** Subgroup of $GL_{2n}(\mathbb{K})$ preserving a non-degenerate skew-symmetric bilinear form. With the standard form, this is the group of matrices A satisfying $A^T J A = J$, where $J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$. All symplectic matrices automatically have determinant 1.
- **Unitary Group (U_n):** This is a real Lie group. It is the subgroup of $GL_n(\mathbb{C})$ preserving a positive definite Hermitian inner product on $\mathbb{C}^n$. The standard choice gives the condition $A^\dagger A = I$, where $A^\dagger = \overline{A^T}$. The **special unitary group** SU_n requires $\det(A) = 1$.

1.3 Constructions: Covering Groups and Quotients

The theory of Lie groups interacts nicely with topology, especially regarding covering spaces.

Proposition 1.5. Let G be a Lie group and $\pi : \tilde{G} \to G$ its universal covering space. There is a unique Lie group structure on $\tilde{G}$ such that π is a morphism of Lie groups. The kernel of π is $\pi_1(G)$, which is a discrete subgroup of the center $Z(\tilde{G})$.

Proposition 1.6. Conversely, if H is a Lie group and $\Gamma \subset Z(H)$ is a discrete central subgroup, then the quotient group $G = H/\Gamma$ has a unique Lie group structure such that the projection map $H \to G$ is a morphism.

Definition 1.7. Two Lie groups are **isogenous** if they have isomorphic universal covers. This means they are locally isomorphic and share the same Lie algebra.

Example 1.8. The group $SU(2)$ is the universal cover of $SO(3)$. The covering map $SU(2) \to SO(3)$ is 2-to-1. Similarly, $SL_2(\mathbb{C})$ is the universal cover of $SO_3(\mathbb{C})$. These are fundamental relationships in geometry and physics.

2 Lecture 8: Lie Algebras and Lie Groups

2.1 Motivation and Definition

The structure of a Lie group near its identity element can be "linearized." This linearization is a vector space with an additional algebraic structure, which captures the essential local properties of the group.

Definition 2.1. The **Lie algebra** of a Lie group G, denoted $\mathfrak{g}$ or $\mathrm{Lie}(G)$, is the tangent space to the manifold G at the identity element e:

$$\mathfrak{g} = T_e G$$

A homomorphism of connected Lie groups $\rho : G \to H$ is completely determined by its differential at the identity, $d\rho_e : \mathfrak{g} \to \mathfrak{h}$. This is the **First Principle** of Lie theory. This raises the question: which linear maps between tangent spaces actually come from group homomorphisms? The answer lies in an additional structure on $\mathfrak{g}$.

For any $g \in G$, conjugation $\Psi_g(h) = ghg^{-1}$ is an automorphism of G that fixes the identity e. Its differential is a linear map $d(\Psi_g)_e : \mathfrak{g} \to \mathfrak{g}$. This gives rise to a representation of the group on its own tangent space:

Definition 2.2. The **adjoint representation** of a Lie group G is the map $Ad : G \to \mathrm{Aut}(\mathfrak{g})$ given by $Ad(g) = d(\Psi_g)_e$.

Since Ad is a map of Lie groups, we can take its differential at the identity to get a linear map from $\mathfrak{g}$ to the Lie algebra of $\mathrm{Aut}(\mathfrak{g})$, which is $\mathrm{End}(\mathfrak{g})$:

Definition 2.3. The **adjoint representation** of a Lie algebra $\mathfrak{g}$ is the map $ad : \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$ given by $ad = d(Ad)_e$. The **Lie bracket** on $\mathfrak{g}$ is defined by this map:

$$[X, Y] := ad(X)(Y)$$

For a matrix group $G \subset GL_n(\mathbb{K})$, this abstract definition simplifies. The Lie algebra $\mathfrak{g}$ is a vector subspace of $M_n(\mathbb{K})$, and the Lie bracket is the commutator of matrices:

$$[X, Y] = XY - YX$$

A linear map $L : \mathfrak{g} \to \mathfrak{h}$ is the differential of a group homomorphism if and only if it preserves the bracket structure: $L([X, Y]) = [L(X), L(Y)]$. This is the **Second Principle** of Lie theory.

Definition 2.4. An abstract **Lie algebra** is a vector space $\mathfrak{g}$ over a field $\mathbb{K}$ equipped with a bilinear map $[\cdot,\cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$, called the Lie bracket, satisfying two axioms for all $X, Y, Z \in \mathfrak{g}$:

1. **Anti-commutativity:** $[X, Y] = -[Y, X]$.
2. **The Jacobi Identity:** $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$.

A **representation** of a Lie algebra $\mathfrak{g}$ on a vector space V is a Lie algebra homomorphism $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$, where $\mathfrak{gl}(V)$ is the Lie algebra of endomorphisms of V with the commutator bracket.

2.2 Examples of Lie Algebras

We find the Lie algebras of the classical groups by considering paths of the form $\gamma(t) = I + tX + O(t^2)$ and enforcing the group conditions by differentiating at $t = 0$.

- $\mathfrak{gl}_n(\mathbb{K}) = M_n(\mathbb{K})$: The Lie algebra of $GL_n(\mathbb{K})$ is the space of all $n \times n$ matrices.
- $\mathfrak{sl}_n(\mathbb{K}) = \{X \in M_n(\mathbb{K}) \mid \mathrm{Tr}(X) = 0\}$: The condition $\det(I + tX) = 1 + t \cdot \mathrm{Tr}(X) + O(t^2) = 1$ implies $\mathrm{Tr}(X) = 0$.
- $\mathfrak{so}_n(\mathbb{K}) = \{X \in M_n(\mathbb{K}) \mid X^T = -X\}$: Differentiating $(I + tX)^T(I + tX) = I$ gives $I + tX^T + tX + O(t^2) = I$, which implies $X^T + X = 0$.
- $\mathfrak{sp}_{2n}(\mathbb{K}) = \{X \in M_{2n}(\mathbb{K}) \mid X^T J + JX = 0\}$: Similarly, differentiating $(I + tX)^T J(I + tX) = J$ gives the condition.
- $\mathfrak{u}_n = \{X \in M_n(\mathbb{C}) \mid X^\dagger = -X\}$: Skew-Hermitian matrices.
- $\mathfrak{su}_n = \{X \in M_n(\mathbb{C}) \mid X^\dagger = -X, \mathrm{Tr}(X) = 0\}$: Traceless skew-Hermitian matrices.

2.3 Tensor Products of Representations

The way tensor products are defined for representations of Lie algebras is different from Lie groups, but is determined by the group definition. If V and W are representations of a Lie group G, then $g(v \otimes w) = g(v) \otimes g(w)$. If we take a path $g(t) = \exp(tX)$, then

$$\begin{aligned} X(v \otimes w) &= \left.\frac{d}{dt}\right|_{t=0} (\exp(tX)v \otimes \exp(tX)w) \\ &= (Xv \otimes w) + (v \otimes Xw) \end{aligned}$$

This is the Leibniz rule. So, for a representation of a Lie algebra $\mathfrak{g}$, the action on a tensor product is $X(v \otimes w) = X(v) \otimes w + v \otimes X(w)$. Similarly, for the dual representation V^*, the action is $(X\phi)(v) = -\phi(Xv)$.

2.4 The Exponential Map

The bridge connecting a Lie algebra back to its Lie group is the exponential map.

Definition 2.5. For $X \in \mathfrak{g}$, there exists a unique Lie group homomorphism $\phi_X : (\mathbb{R}, +) \to G$ whose differential at the origin is the map $1 \mapsto X$. This is called a **one-parameter subgroup**. The **exponential map** is then defined as $\exp(X) = \phi_X(1)$.

For matrix Lie groups, this abstract definition coincides with the standard matrix exponential series:

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!} = I + X + \frac{X^2}{2} + \frac{X^3}{6} + \dots$$

The exponential map has two foundational properties that establish the deep correspondence between Lie groups and Lie algebras:

- It defines a local diffeomorphism between a neighborhood of $0 \in \mathfrak{g}$ and a neighborhood of $e \in G$.
- It is natural with respect to morphisms: if $\phi : G \to H$ is a morphism of Lie groups, it induces a morphism of Lie algebras $d\phi : \mathfrak{g} \to \mathfrak{h}$, and the following diagram commutes:

$$
\begin{array}{ccc}
\mathfrak{g} & \xrightarrow{d\phi} & \mathfrak{h} \\
\downarrow \exp & & \downarrow \exp \\
G & \xrightarrow{\phi} & H
\end{array}
$$

This correspondence is profound. A homomorphism between connected Lie groups is entirely determined by the corresponding map of their Lie algebras. The local group structure is fully encoded in the Lie bracket. The full group multiplication law can be recovered from the Lie algebra via the **Baker-Campbell-Hausdorff formula**:

$$\log(\exp(X)\exp(Y)) = X + Y + \frac{1}{2}[X,Y] + \frac{1}{12}([X,[X,Y]] - [Y,[X,Y]]) + \dots$$

where all terms are expressed using only Lie brackets.

Proposition 2.6 (**The Lie Correspondence**). For a Lie group G, there is a one-to-one correspondence between its connected Lie subgroups and the Lie subalgebras of its Lie algebra $\mathfrak{g}$. Normal subgroups correspond to ideals.

This correspondence is the cornerstone of representation theory for Lie groups. It allows us to translate problems about group representations (which involve smooth manifolds) into problems about Lie algebra representations (which are problems of linear algebra).

Conclusion

Lectures 7 and 8 establish the fundamental objects of our study. A Lie group is a smooth manifold with a compatible group structure. To each Lie group, we can associate its linearization at the identity, the Lie algebra, which is a vector space equipped with the Lie bracket. The exponential map provides the bridge from the algebra back to the group. This powerful dictionary allows us to study the intricate world of continuous symmetries using the robust tools of algebra.